



CLOUD COMPUTING

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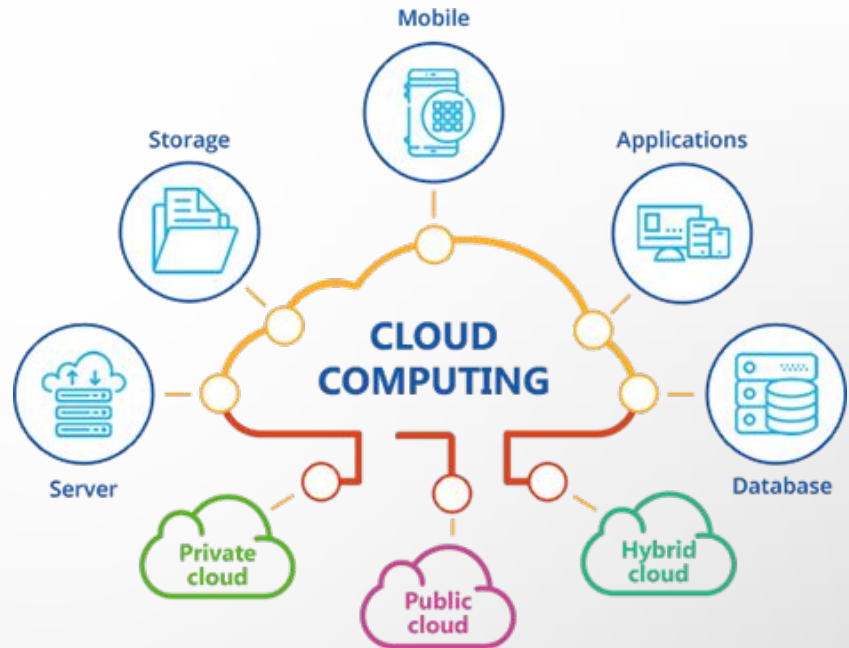
**Department of CSE- AIML & IoT
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UNIT-I:

- **CLOUD COMPUTING AT A GLANCE:** THE VISION OF CLOUD COMPUTING, DEFINING A CLOUD ,A CLOSER LOOK, THE CLOUD COMPUTING REFERENCE MODEL ,CHARACTERISTICS AND BENEFITS, CHALLENGES AHEAD
- **HISTORICAL DEVELOPMENTS:** DISTRIBUTED SYSTEMS, VIRTUALIZATION, WEB2.0 SERVICE- ORIENTED COMPUTING UTILITY-ORIENTED COMPUTING
- **BUILDING CLOUD COMPUTING ENVIRONMENTS:** APPLICATION DEVELOPMENT INFRASTRUCTURE AND SYSTEM DEVELOPMENT, COMPUTING PLATFORMS AND TECHNOLOGIES.

INTRODUCTION

Computing is being transformed into A model consisting of services that are commoditized and delivered in A manner similar to utilities such as water, electricity, gas, and telephony.



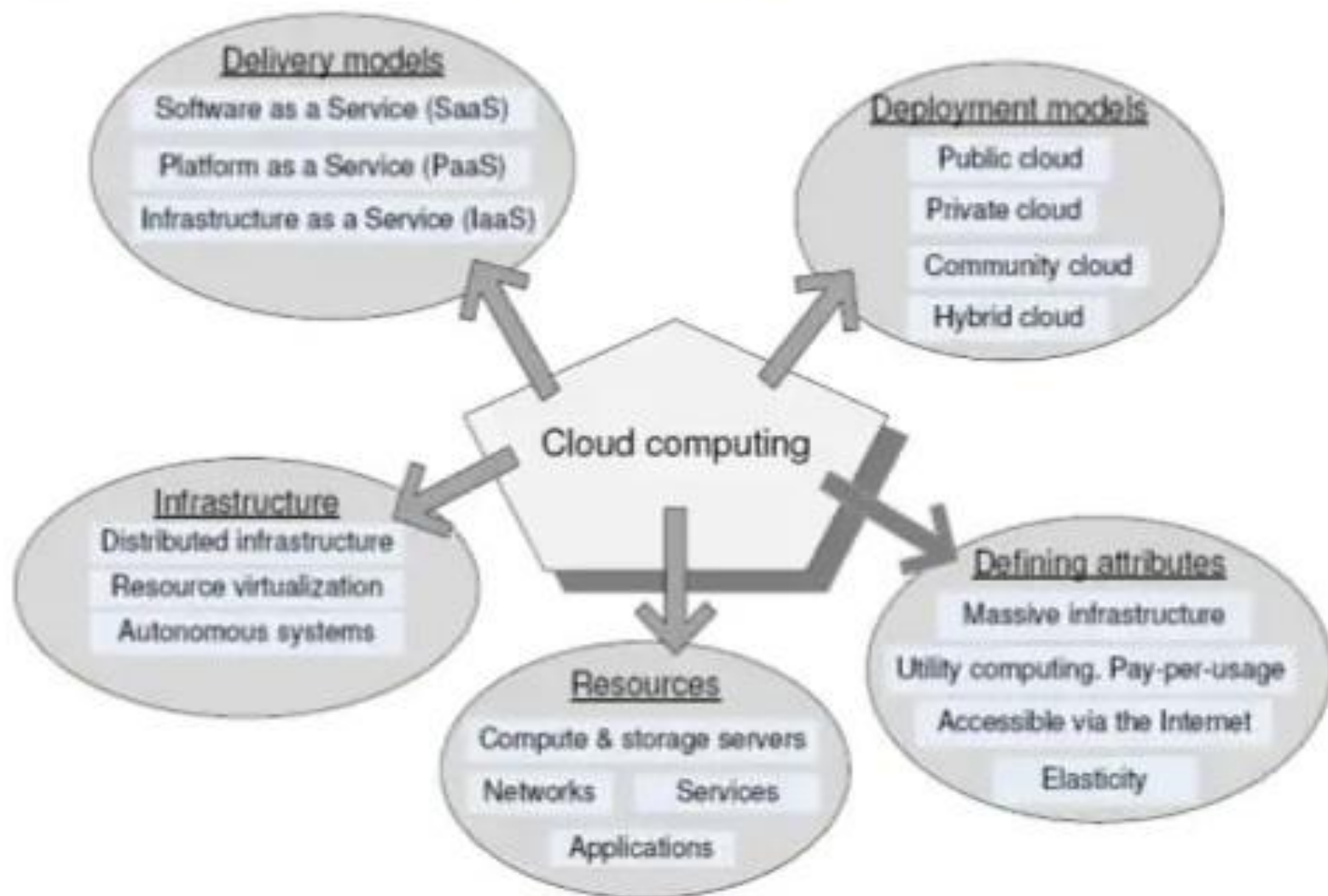
In such a model, users access services based on their requirements, regardless of where the services are hosted.

Cloud computing is the most recent emerging paradigm promising to turn the vision of “computing utilities” into a reality.

CLOUD COMPUTING AT A GLANCE

- This vision of computing utilities based on A service-provisioning model anticipated the massive transformation of the entire computing industry in the 21st century.
- Computing services will be readily available on demand, just as other utility services such as water, electricity, telephone, and gas are available in today's society.
- Users(consumers)need to pay providers only when they access the computing services.
- In addition, consumers no longer need to invest heavily or encounter difficulties in building and maintaining complex it infrastructure.

Cloud Computing: In a Glance



THE VISION OF CLOUD COMPUTING

- Cloud computing is a technological advancement that focuses on the way we design computing systems, develop applications, and leverage existing services for building software.
- It is based on the concept of dynamic provisioning, which is applied not only to services but also to compute capability, storage, networking, and information technology(it) infrastructure in general.
- Today, anyone with a credit card can subscribe to cloud services and deploy and configure servers for an application in hours, growing and shrinking the infrastructure serving its application according to the demand, and paying only for the time these resources have been used.

THE VISION OF CLOUD COMPUTING

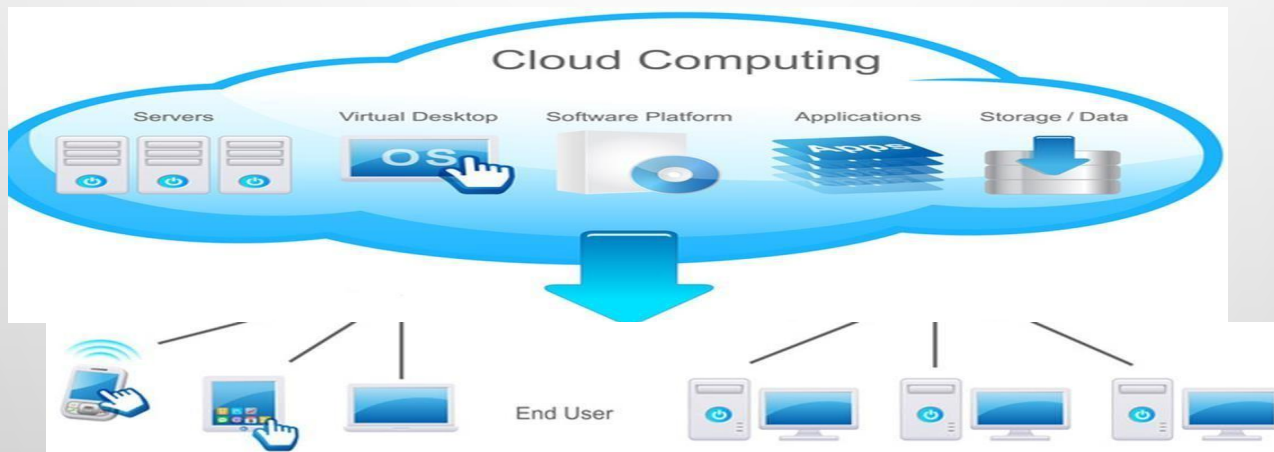


DEFINING A CLOUD

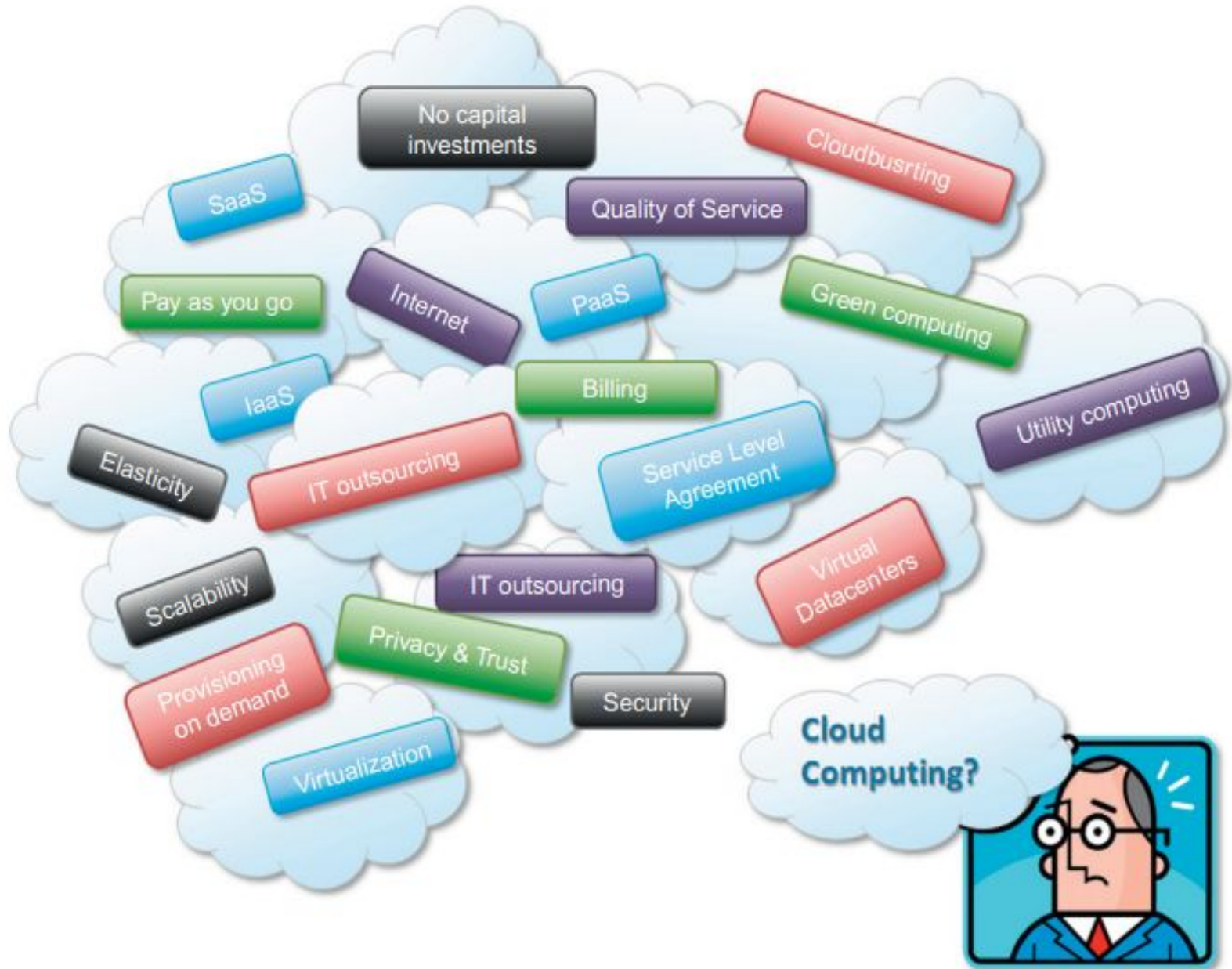
- Cloud computing has become a popular buzzword; it has been widely used to refer to different technologies, services, and concepts.
- It is often associated with virtualized infrastructure or hardware on demand, utility computing, it outsourcing, platform and software as a service, and many other things that now are the focus of the it industry.
- The term cloud has historically been used in the telecommunications industry as an abstraction of the network in system diagrams
- Cloud computing refers to both the applications delivered as services over the internet and the hardware and system software in the datacentres that provide those services.

DEFINITION OF CLOUD COMPUTING

- Cloud computing is a **service provisioning technique** where **computing resources** like hardware such as **servers and storage devices, software's** and **complete platform for developing applications** are provided as a **service** by the **cloud providers** to the **customers**.



- Customers can use these resources as and when needed, can increase or decrease resource capacities dynamically according to their requirements and pay only for how much the resource were used.



Cloud Computing Technologies, Concepts and Ideas

- *Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (ex: Networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.*

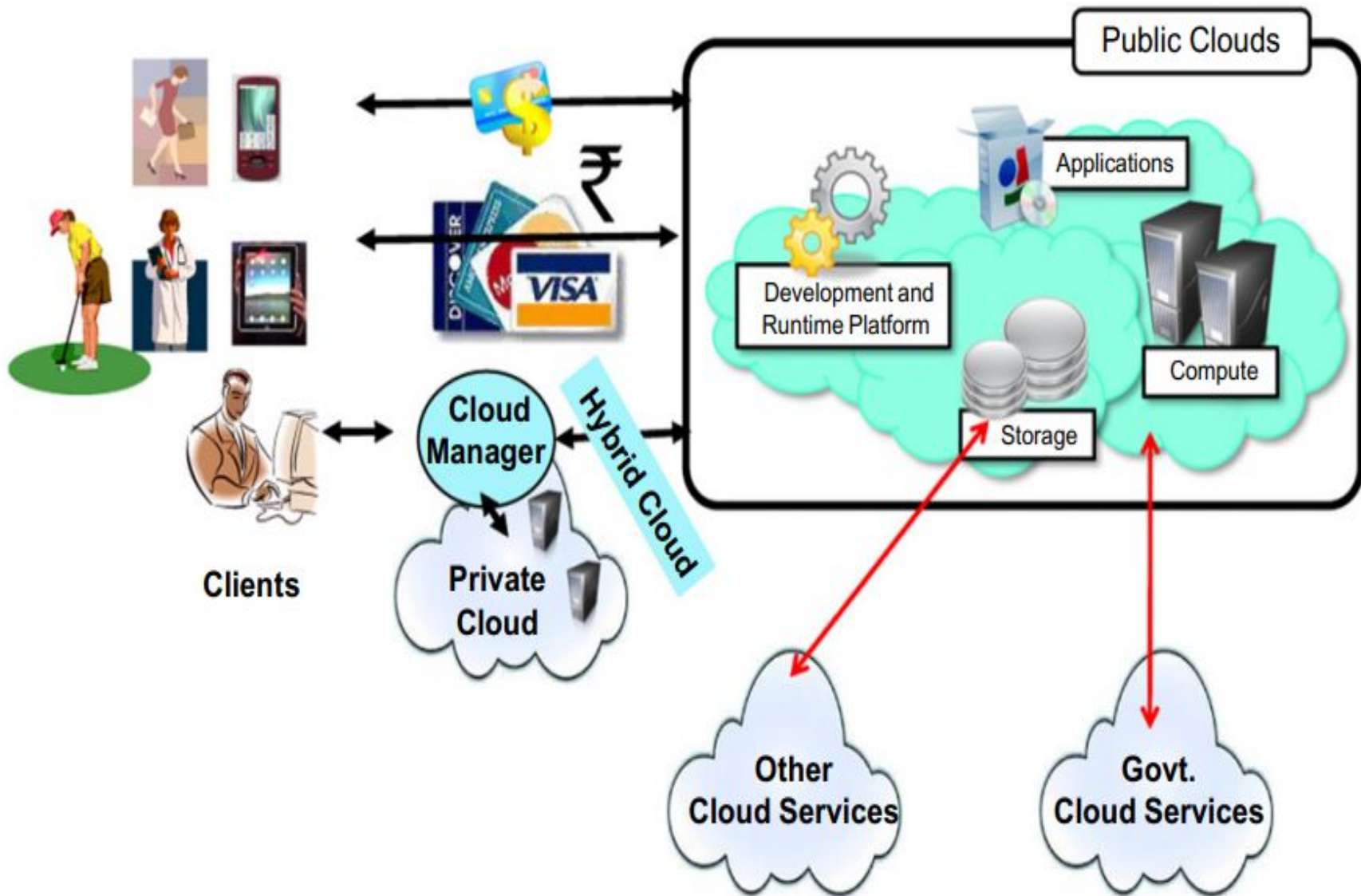
We can define three criteria to discriminate whether a service is delivered in the cloud computing style:

- The service is accessible via a web browser (non-proprietary) or a web services application programming interface (API).
- Zero capital expenditure is necessary to get started.
- You pay only for what you use as you use it.

A CLOSER LOOK

- Cloud computing is helping enterprises, governments, public and private institutions, and research organizations shape more effective and demand-driven computing systems.
- ❖ **Large enterprises can offload some of their activities to cloud-based systems:** Recently, the New York Times has converted its digital library about past editions into web friendly formats- EC2,S3 (it performs in 36 hours)
- ❖ **Small enterprises and start-ups can afford to translate their ideas into business results more quickly, without excessive up-front costs:** Animoto is a company that creates videos out of images, music, and video fragments submitted by users.(Animoto –no servser-takes from AWS)
- ❖ **System developers can concentrate on the business logic rather than dealing with the complexity of infrastructure management and scalability:** Little Fluffy Toys is a company in London that has developed a widget providing users with information about nearby rental bicycle services.-(widget computing needs on google AppEngine)
- ❖ **End users can have their documents accessible from everywhere and any device :** Apple icloud is a service that allows users to have their documents stored in the cloud and access from the device they connect to it.

A BIRD'S-EYE VIEW OF CLOUD COMPUTING.



CLOUD DEPLOYMENT MODELS

Public/Internet Clouds

***Third-party, multitenant cloud infrastructure and services**

***Available on a subscription basis to all**



Private/Enterprise Clouds

***A public cloud model within a company's own datacenter/infrastructure for internal and/or partners' use**



Hybrid/Inter Clouds

*** Mixed use of private and public clouds; leasing public cloud services when private cloud capacity is insufficient**



THE CLOUD COMPUTING REFERENCE MODEL

- It has the capability to deliver, on demand, A variety of IT services that are quite diverse from each other.
- This variety creates different perceptions of what cloud computing is among users.
- Despite this lack of uniformity, it is possible to classify cloud computing services offerings into three major categories:
 - **INFRASTRUCTURE-AS-A-SERVICE (IAAS)**
 - **PLATFORM-AS-A-SERVICE (PAAS) AND**
 - **SOFTWARE-AS-A-SERVICE (SAAS).**

Web 2.0
Interfaces

Software as a Service



Examples : Google Documents, Facebook, Flickr, Salesforce

End-user applications

Scientific applications

Office automation, photo editing,

CRM, and social networking

Platform as a Service



Examples : Windows Azure, Hadoop, Google AppEngine, Aneka

Runtime environment for applications

Development and data processing platforms

Infrastructure as a Service



Examples : Amazon EC2, S3, Rightscale, vCloud

Virtualized servers

Storage and networking



CHARACTERISTICS AND BENEFITS

- Cloud Computing Has Some Interesting Characteristics that Bring Benefits to Both Cloud Service Consumers (Cscs) And Cloud Service Providers (Csps). These Characteristics Are:
 - No Up-front Commitments
 - On-demand Access
 - Nice Pricing
 - Simplified Application Acceleration And Scalability
 - Efficient Resource Allocation
 - Energy Efficiency
 - Seamless Creation And Use Of Third-party Services

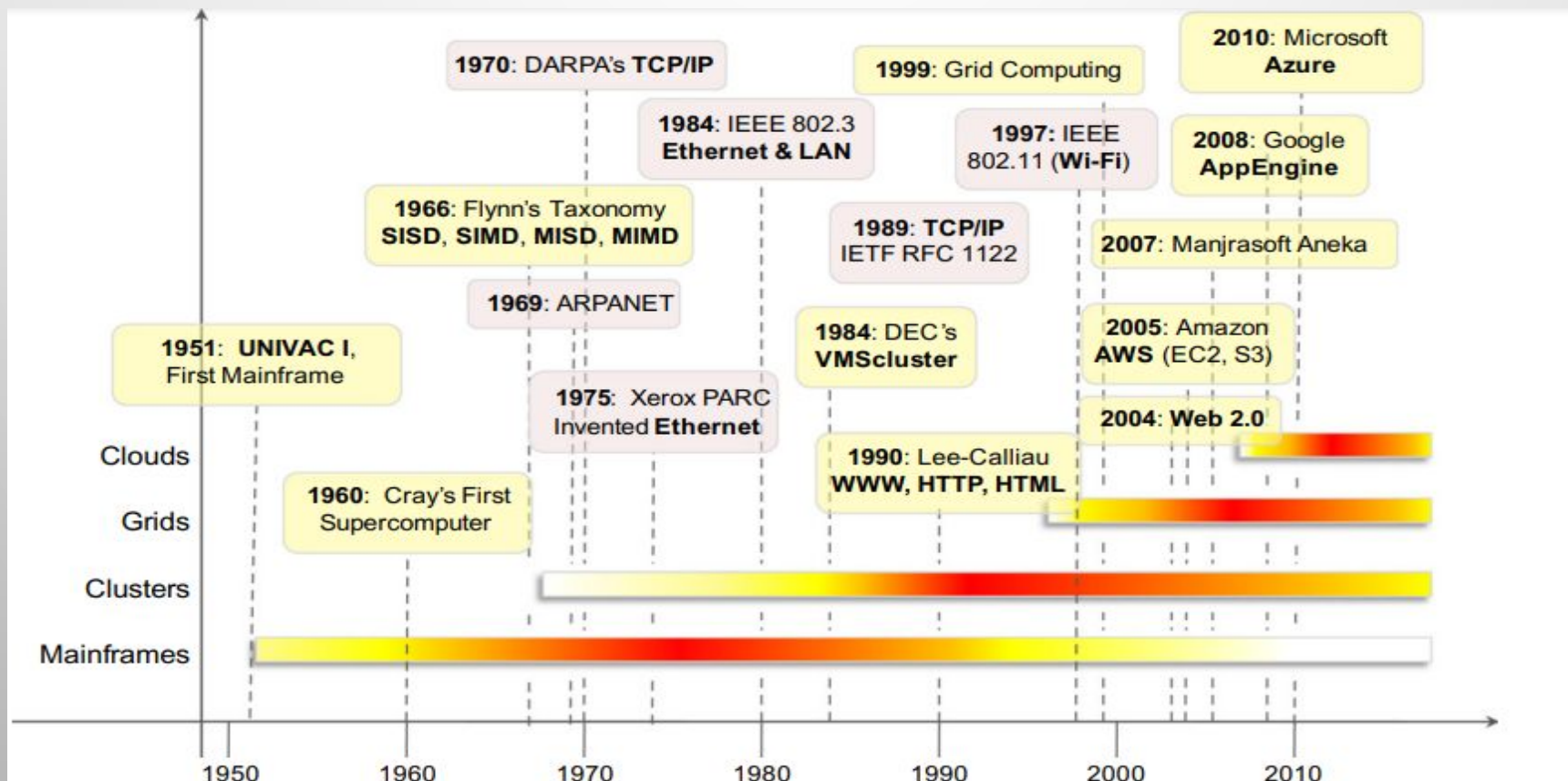
CHALLENGES AHEAD

- As any new technology develops and becomes popular, new issues have to be faced.
- New, interesting problems and challenges are regularly being posed to the cloud community, including its practitioners, managers, governments, and regulators.
- Which are related to configuration, networking, and sizing of cloud computing systems, a new set of challenges concerning the dynamic provisioning of cloud computing services and resources arises.
- Technical challenges also arise for cloud service providers for the management of large computing infrastructures and the use of virtualization technologies .
- Security in terms of confidentiality, secrecy, and protection of data in a cloud environment is another important challenge.
- The confidentiality of data and compliance to security standards, which give a minimum guarantee on the treatment of information on cloud computing systems (cryptographic).
- Different legislation about privacy in different countries may potentially create disputes as to the rights that third parties (including government agencies) have to your data.

Historical Developments: Distributed systems, Virtualization, Web2.0, Service- oriented computing , Utility-oriented computing

HISTORICAL DEVELOPMENTS

The idea of renting computing services by leveraging large distributed computing facilities has been around for long time. It dates back to the days of the mainframes in the early 1950s. From there on, technology has evolved and been refined. We can review five core technologies that played an important role in the realization of cloud computing. These technologies are **distributed systems, virtualization, Web 2.0, service orientation, and utility computing.**



DISTRIBUTED SYSTEMS:

Clouds are essentially large distributed computing facilities that make available their services to third parties on demand.

“A distributed system is a collection of independent computers that appears to its users as a single coherent system.”

The primary purpose of distributed systems is to share resources and utilize them better.

Distributed systems often exhibit other properties such as heterogeneity, openness, scalability, transparency, concurrency, continuous availability, and independent failures.

Three major milestones have led to cloud computing:

MAINFRAME COMPUTING

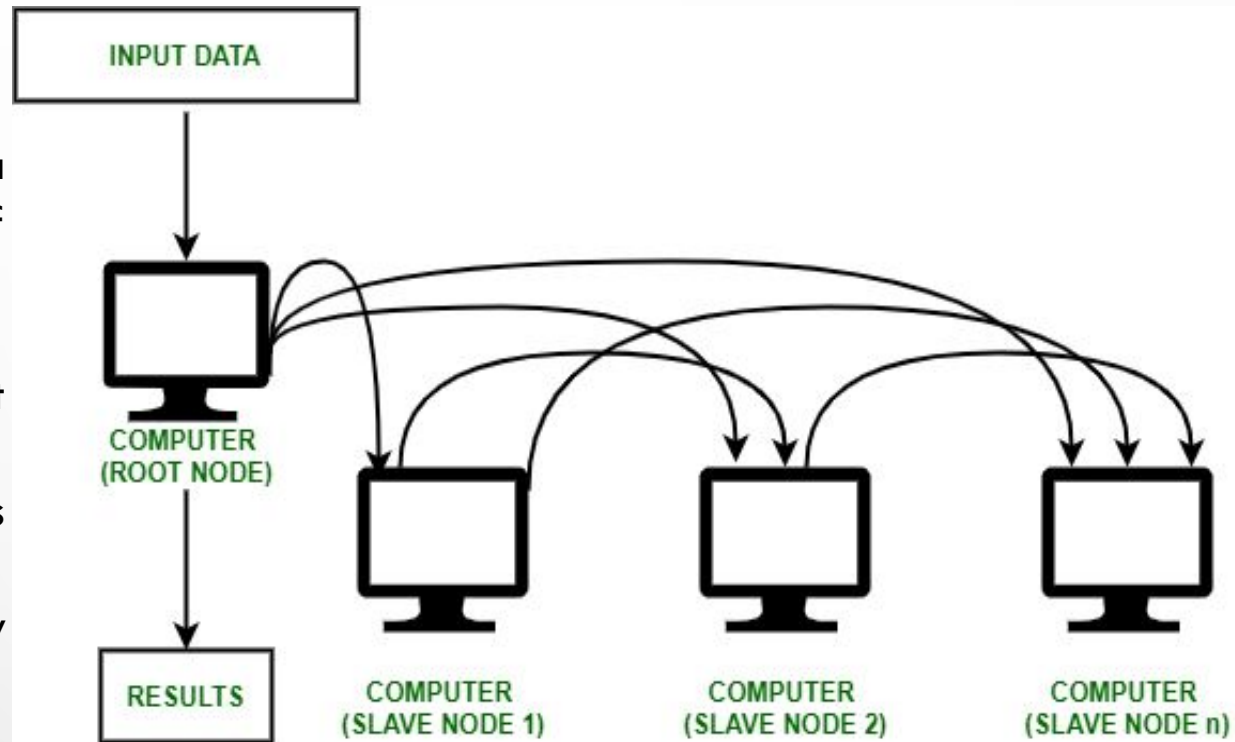
CLUSTER COMPUTING AND

GRID COMPUTING

MAINFRAMES

- large computational facilities leveraging multiple processing units.
- Mainframes were powerful, highly reliable computers specialized for large data movement and massive input/output (I/O) operations.
- used by large organizations for bulk data processing tasks such as online transactions, enterprise resource planning, and other operations involving the processing of significant amounts of data.
- One of the most attractive features of mainframes was the ability to be highly reliable computers that were “always on” and capable of tolerating failures transparently. No system shutdown was required to replace failed components, and the system could work without interruption. Batch processing was the main application of mainframes.
- online banking, airline ticket booking, supermarket and telecoms, and government services





Cluster computing started as a low-cost alternative to the use of mainframes and supercomputers.

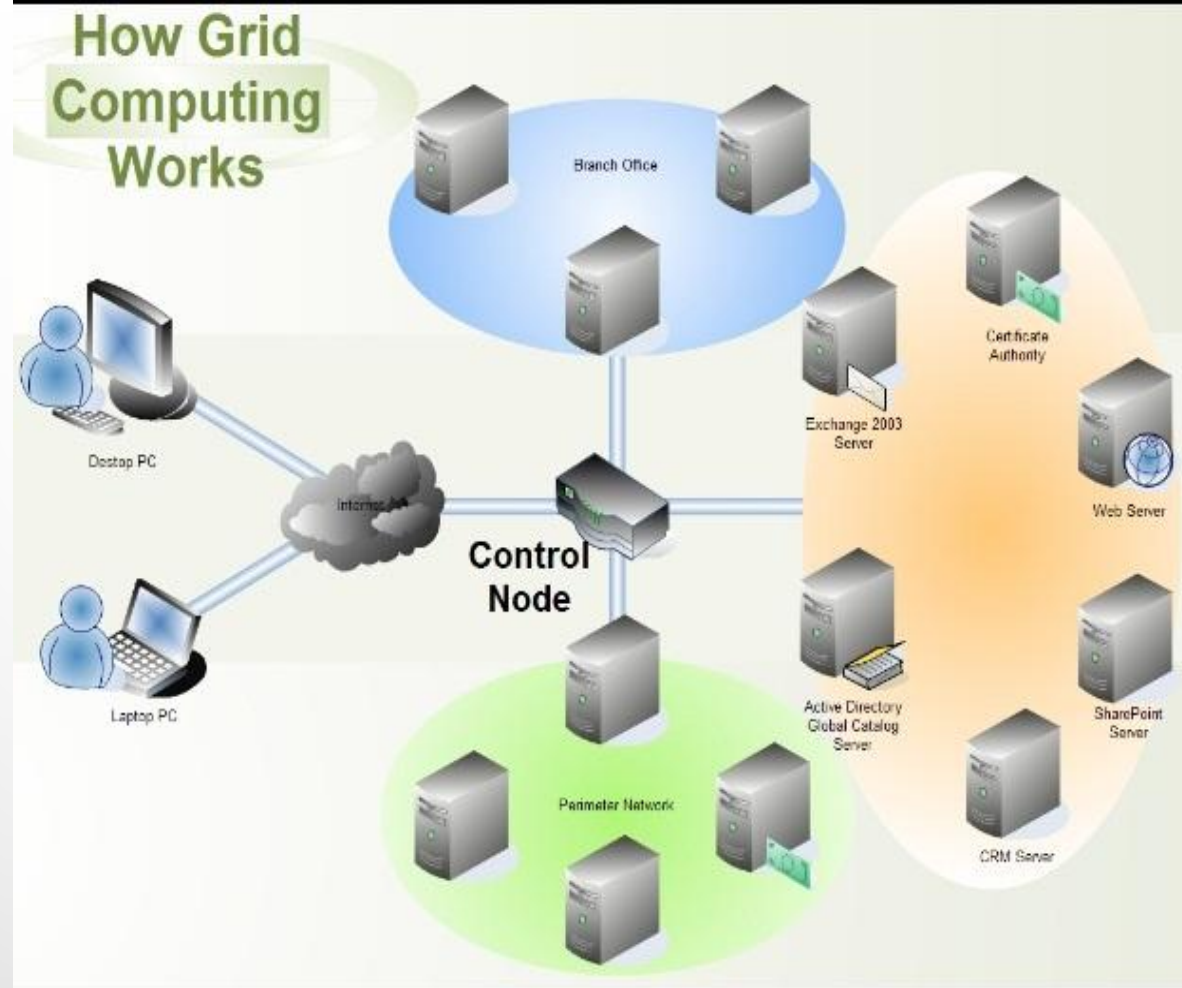
The technology advancement that created faster and more powerful mainframes and supercomputers eventually generated an increased availability of cheap commodity machines as a side effect.

- These machines could then be connected by a high-bandwidth network and controlled by specific software tools that manage them as a single system.
- Starting in the 1980s, clusters became the standard technology for parallel and high-performance computing. Built by commodity machines, they were cheaper than mainframes and made high-performance computing available to a large number of groups, including universities and small research labs
- One of the attractive features of clusters was that the computational power of commodity machines could be leveraged to solve problems that were previously manageable only on expensive supercomputers.

Grid computing appeared in the early 1990s as an evolution of cluster computing.

Grid computing proposed a new approach to access large computational power, huge storage facilities, and a variety of services.

Grids initially developed as aggregations of geographically dispersed clusters by means of internet connections.



Different from a “large cluster,” a computing grid was a dynamic aggregation of heterogeneous computing nodes, and its scale was nationwide or even worldwide.

VIRTUALIZATION:

- Virtualization is another core technology for cloud computing. It encompasses a collection of solutions allowing the abstraction of some of the fundamental elements for computing, such as hardware, runtime environments, storage, and networking. This is particularly true for Solutions that provide IT infrastructure on demand. Virtualization confers that degree of customization and control that makes cloud computing appealing for users and, at the same time, sustainable for cloud services providers.
- Virtualization is essentially a technology that allows creation of different computing environments. These environments are called virtual because they simulate the interface that is expected By a guest. The most common example of virtualization is hardware virtualization. This technology allows simulating the hardware interface expected by an operating system.
- Hardware virtualization Allows the coexistence of different software stacks on top of the same hardware. These stacks are contained inside virtual machine instances, which operate in complete isolation from each other.
- Amazon EC2, rightscale, vmware vcloud, and others. Together with hardware virtualization, storage and network virtualization complete the range of technologies for the emulation of IT infrastructure.

WEB 2.0:

- The Web is the primary interface through which cloud computing delivers its services. At present, the Web encompasses a set of technologies and services that facilitate interactive information sharing, collaboration, user-centered design, and application composition.
- This evolution has transformed the Web into a rich platform for application development and is known as Web 2.0.
- Web 2.0 brings interactivity and flexibility into Web pages, providing enhanced user experience by gaining Web-based access to all the functions that are normally found in desktop applications.
- These capabilities are obtained by integrating a collection of standards and technologies such as XML, Asynchronous JavaScript and XML (AJAX), Web Services, and others. These technologies allow us to build applications leveraging the contribution of users, who now become providers of content.
- Examples of Web 2.0 applications are Google Documents, Google Maps, Flickr, Facebook, Twitter, YouTube, Blogger, and Wikipedia. In particular, social networking Websites take the biggest advantage of Web 2.0.

Service-oriented computing:

- Service orientation is the core reference model for cloud computing systems. This approach adopts the concept of services as the main building blocks of application and system development.
- Service-oriented computing (soc) supports the development of rapid, low-cost, flexible, interoperable, and evolvable applications and systems
- A service is supposed to be loosely coupled, reusable, programming language independent, and location transparent. Loose coupling allows services to serve different scenarios more easily and makes them reusable.
- Service-oriented computing introduces and diffuses two important concepts, which are also fundamental to cloud computing: quality of service (qos) and software-as-a-service (saas).
- Quality of service (qos) identifies a set of functional and nonfunctional attributes that can be used to evaluate the behavior of a service from different perspectives. These could be performance metrics such as response time, or security attributes, transactional integrity, reliability, scalability, and availability
- Software-as-a-service introduces a new delivery model for applications. The term has been inherited from the world of application service providers (asps), which deliver software services-based solutions across the wide area network from a central datacenter and make them available on a subscription or rental basis.

Utility-oriented computing

- Utility computing is a vision of computing that defines a service-provisioning model for compute services in which resources such as storage, compute power, applications, and infrastructure are packaged and offered on a pay-per-use basis. The idea of providing computing as a utility like natural gas, water, power, and telephone connection has a long history but has become a reality today with the advent of cloud computing.

Building cloud computing environments:

Application development, Infrastructure and
System development, Computing platforms and
technologies.

BUILDING CLOUD COMPUTING ENVIRONMENTS

- The creation of cloud computing environments encompasses both the development of applications and systems that leverage cloud computing solutions and the creation of frameworks, platforms, and infrastructures delivering cloud computing services.
- **Application development:** Applications that leverage cloud computing benefit from its capability to dynamically scale on Demand. One class of applications that takes the biggest advantage of this feature is that of web Applications. Their performance is mostly influenced by the workload generated by varying user demands. With the diffusion of web 2.0 technologies, the web has become a platform for developing rich and complex applications, including enterprise applications that now leverage the internet as the preferred channel for service delivery and user interaction.
- Cloud computing provides a solution for on-demand and dynamic scaling across the entire stack Of computing. This is achieved by
 - (a) providing methods for renting compute power, storage, and Networking;
 - (b) offering runtime environments designed for scalability and dynamic sizing;
 - (c) providing application services that mimic the behavior of desktop applications but that are Completely hosted and managed on the provider side.

INFRASTRUCTURE AND SYSTEM DEVELOPMENT :

Distributed computing, virtualization, service orientation, and web 2.0 form the core technologies enabling the provisioning of cloud services from anywhere on the globe. Developing applications and systems that leverage the cloud requires knowledge across all these technologies. Moreover, new challenges need to be addressed from design and development standpoints.

- Distributed computing is a foundational model for cloud computing because cloud systems are distributed systems. besides administrative tasks mostly connected to the accessibility of Resources in the cloud, the extreme dynamism of cloud systems—where new nodes and services are provisioned on demand—constitutes the major challenge for engineers and developers. This Characteristic is pretty peculiar to cloud computing solutions. Infrastructure-as-a-service solutions provide the capabilities to add and remove resources, but it is up to those who deploy systems on this scalable infrastructure to make use of such opportunities with wisdom and effectiveness. Platform-as-a-service solutions embed into their core offering algorithms and rules that control the provisioning process and the lease of resources.
- Cloud computing is often summarized with the acronym **Xaas—everything-as-a-service**—that clearly underlines the central role of service orientation.

COMPUTING PLATFORMS AND TECHNOLOGIES

- Development of a cloud computing application happens by leveraging platforms and frameworks that provide different types of services, from the bare-metal infrastructure to customizable applications serving specific purposes.

1. AMAZON WEB SERVICES (AWS) :

- AWS offers comprehensive cloud IaaS services ranging from virtual compute, storage, and Networking to complete computing stacks. AWS is mostly known for its compute and storage-on Demand services, namely elastic compute cloud (EC2) and simple storage service (S3).
- EC2 Provides users with customizable virtual hardware that can be used as the base infrastructure for deploying computing systems on the cloud. It is possible to choose from a large variety of virtual Hardware configurations, including GPU and cluster instances.
- EC2 instances are deployed either by using the AWS console, which is a comprehensive web portal for accessing AWS services, or by using the web services API available for several programming languages.
- EC2 Also provides the capability to save a specific running instance as an image, thus allowing users to create their own templates for deploying systems. These templates are stored into S3 that delivers persistent Storage on demand.
- S3 is organized into buckets; these are containers of objects that are stored in binary form and can be enriched with attributes. Users can store objects of any size, from simple Files to entire disk images, and have them accessible from everywhere.

2. GOOGLE APP ENGINE:

- GOOGLE APP ENGINE Is a scalable runtime environment mostly devoted to executing web applications. These take advantage of the large computing infrastructure of google to dynamically scale as the demand varies over time.
- App engine provides both a secure execution environment and a collection of services that simplify the development of scalable and high-performance web applications. These services include in-memory caching, scalable data store, job queues, messaging tasks. Developers can build and test applications on their own machines using the appengine.
- Software development kit (SDK), which replicates the production runtime environment and helps test and profile applications.
- Once development is complete, developers can easily migrate their application to app engine, set quotas to contain the costs generated, and make the application available to the world.

3 MICROSOFT AZURE:

- MICROSOFT AZURE Is a cloud operating system and a platform for developing applications in the Cloud. It provides a scalable runtime environment for web applications and distributed applications in general. Applications in azure are organized around the concept of roles, which identify a distribution unit for applications and embody the application's logic.
- Currently, there are three types of Role: web role, worker role, and virtual machine role.
- The web role is designed to host a web Application, the worker role is a more generic container of applications and can be used to perform Workload processing, and the virtual machine role provides a virtual environment in which the Computing stack can be fully customized, including the operating systems.
- Besides roles, azure Provides a set of additional services that complement application execution, such as support for Storage (relational data and blobs), networking, caching, content delivery, and others.

4 HADOOP

- Apache hadoop is an open-source framework that is suited for processing large data sets on commodity hardware. Hadoop is an implementation of mapreduce, an application programming model developed by google, which provides two fundamental operations for data processing: map and Reduce.
- The former transforms and synthesizes the input data provided by the user, the latter aggregates the output obtained by the map operations. Hadoop provides the runtime environment, and developers need only provide the input data and specify the map and reduce functions that need to be executed.
- Yahoo!, The sponsor of the apache hadoop project, has put considerable effort into transforming the project into an enterprise-ready cloud computing platform for data processing.

5: FORCE.COM AND SALESFORCE.COM

- Force.Com is a cloud computing platform for developing social enterprise applications. The platform is the basis for salesforce.Com, a software-as-a-service solution for customer relationship Management.
- Force.Com allows developers to create applications by composing ready-to-use Blocks, a complete set of components supporting all the activities of an enterprise are available. It is also possible to develop your own components or integrate those available in app exchange into your applications. The platform provides complete support for developing applications, from the design of the data layout to the definition of business rules and workflows and the definition of the User interface.
- The force.Com platform is completely hosted on the cloud and provides complete access to its functionalities and those implemented in the hosted applications through web services technologies.

6 MANJRASOFT ANEKA :

- Manjrasoft aneka [165] is a cloud application platform for rapid creation of scalable applications and their deployment on various types of clouds in a seamless and elastic manner. It supports a collection of programming abstractions for developing applications and a distributed runtime environment that can be deployed on heterogeneous hardware (clusters, networked desktop computers, and Cloud resources).
- Developers can choose different abstractions to design their application: tasks, distributed threads, and map-reduce. These applications are then executed on the distributed Service-oriented runtime environment, which can dynamically integrate additional resource on Demand.
- The service-oriented architecture of the runtime has a great degree of flexibility and simplifies the integration of new features, such as abstraction of a new programming model and associated execution management environment.
- Services manage most of the activities happening at Runtime: scheduling, execution, accounting, billing, storage, and quality of service.
- These platforms are key examples of technologies available for cloud computing. They mostly fall into the three major market segments identified in the reference model: infrastructure-as-a-service, Platform-as-a-service, and software-as-a-service.